

Mitchelle Ayanda Mzolo

Full-Stack Engineer | Cloud Infrastructure and Document Intelligence

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Profile

Full-stack engineer who builds and secures the systems that move money, documents and messages between companies: bank submission for a multi-tenant mortgage platform, customer conversations and executive reporting for an internet provider, and pricing logic for a travel booking platform. Strongest at finding the failures that do not announce themselves, proving the cause from production evidence, and fixing it once at the point every caller passes through. Writes the post-mortem and the runbook afterwards.

Technical skills

Backend and platform: PHP, Laravel, REST APIs, webhooks, SOAP with WS-Security, queues, multi-tenancy

Data: MySQL, PostgreSQL (materialised views, plpgsql, row level security), Supabase, Redis, MinIO, MeiliSearch

Security: multi-tenant MFA and OTP, WAF and rate limiting, penetration test remediation, audit logging, secure file handling

Testing and delivery: Pest, PHPUnit, Cypress, GitLab CI, GitHub Actions

Frontend: Vue.js, Inertia.js, Ziggy, TypeScript, Tailwind, Blade

Cloud and infrastructure: AWS (Transfer Family, S3, IAM, VPC, Lambda, CloudWatch), Redshift and DMS, Docker, Linux, NGINX

Document intelligence: Claude on Amazon Bedrock, PaddleOCR, AWS Textract, Python and FastAPI

Experience

Junior Software Developer, Blue Bean Software

November 2025 to present

Multi-tenant mortgage origination platform

- Lead developer and platform owner for a platform serving multiple white-label partners; primary author of 828 of the codebase's 968 commits.
- Own the bank submission integration. Designed a vendor cutover keyed on submission time so applications already in flight completed on their original vendor and nothing had to be resubmitted; later onboarding a new lender became configuration rather than code.
- Caught a routing defect in that cutover by auditing production logs, identified the cause from a split that was deterministic by creation date, fixed it atomically with a backfill, and routed seven further call sites through a single resolver.
- Built document intelligence now in production: a self-hosted PaddleOCR service so client documents never leave the network, with Claude on Amazon Bedrock for structured extraction. Chose the model per document type from measured reliability, and pinned field-hygiene guards to a sweep of over 2,000 real documents.
- Implemented per-tenant MFA toggling and OTP verification across every login surface; configured WAF rules and rate limiting for production; coordinated and remediated penetration test findings.
- Designed and documented an AWS Transfer Family SFTP architecture with S3, IAM roles and CloudWatch logging, and onboarded partners through it with schema validation and scoped access.
- Resolved data integrity issues including duplicate reference numbers and financial calculation inconsistencies; troubleshooted Redshift and DMS synchronisation; upgraded vulnerable NGINX versions and closed sensitive file exposure.

Internet provider

- Built a WhatsApp payment recovery bot end to end as sole author (127 commits) in Laravel, Inertia and Vue 3 on the WhatsApp Cloud API, with tests for out-of-order webhooks, replies on closed conversations, unknown senders,

and records disagreeing with the provider.

- Built an executive and team KPI dashboard (182 commits) in Vue 3, TypeScript and Supabase with CASL permissions: acquisition funnel, product mix, early-life churn and cohort retention. Materialised the views behind it when reports began timing out.
- Worked on the commission system: commission and payout logic, a self-service campaign simulator, order import tooling, and duplicate order prevention.

Full-Stack Developer, Hit The Road Today

August 2024 to July 2026

- Sole author of the promotion and discount engine for a travel booking platform: percentage and cash codes with geographic, product, day-of-week, minimum cart and first-booking rules, modelled as the code, the rule and the redemption.
- Handled refund recalculation when a trip is modified after a discount is applied, and wrote the business logic specification stakeholders signed off from.
- Developed and maintained Laravel and Vue.js platform features; handled production debugging, backend optimisation and data accuracy work in Docker environments with Redis and MinIO; supported deployment cycles and UAT.

Junior Web Developer, Umuzi Academy

August 2023 to August 2024

- Built frontend applications in HTML, CSS and JavaScript, and REST APIs in Node.js with PostgreSQL, in agile teams.

Projects and leadership

- **BillWell** (billwell.netlify.app): invoice generator that renders a styled A4 PDF from a form, storing business details in the browser only. Loads a slimmed Chromium build on serverless hosting where the bundled one will not fit, with a smoke test that verifies the output is a real PDF.
- **ModernTsonga** (modernthsonga.netlify.app): a reference for Xitsonga phrases, proverbs and oral tradition.
- **Mentor, Gaming 4 Girls Hackathon** (Umuzi and UNICEF, 2023): mentored a team to second place.

Education and certification

- AWS Certified Cloud Practitioner (CLF-C02), 2026
- University of South Africa, National Diploma: Information Technology, 2025 to present
- Umuzi Academy, National Certificate: IT Systems Development (NQF 5), 2023 to 2024
- Tshwane University of Technology, National Diploma: IT (incomplete)